

Highway expansion and urban sprawl in the Jakarta Metropolitan Area

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ABSTRACT

Transport development is widely recognized as one of the major drivers in shaping urban forms. While recent literature has documented the urban expansion effect of transport networks between cities, little is known about the effect within larger metropolitan areas. This study aims to find the causal relationships between highway expansion and two different aspects of urban development: urban expansion and urban sprawl in the Jakarta Metropolitan Area (JMA). To obtain the causal effect, we employ historic transport infrastructure as instrumental variables. Our results show that improvement in highway access increases urban sprawl within the JMA, particularly within 40–50 km radius from the city centre. The results also confirm the existence of highway-led urban expansion in several districts within the JMA over the last three decades. This paper adds a piece of enticing evidence that highway expansion leads to a sprawling development of urban areas even within a large metropolitan area.

1. Introduction

Transport networks are closely linked to urban development. The classic monocentric model suggests that people began to move away from city centres as commuting costs fell, causing urban areas to expand (Glaeser and Kahn, 2003). This process promotes faster employment and population growth in fringe urban areas, creating suburbanisation (Baum-Snow, 2007; Garcia-López et al., 2015; Garcia-López, 2012; Yudhistira et al., 2019). While this process is discussed extensively in relation to the outward growth (expansion) of urban areas in developed countries, less is known about how the transport network affects urban land use within metropolitan areas in developing countries, where car dependency is a relatively new phenomenon. Further, these metropolitan areas are typically much larger and grow much faster than their counterparts in developed countries. Such extensive, polycentric urban agglomerations are characterised by complex patterns in which high- and low-density developments alternate at relatively short distances. Each development may have its own relationship with improvements in transport infrastructure. Furthermore, low-density developments associated with the advent of a more affluent middle class (Winarso et al.,

2015), which are commonly referred to as urban sprawl (Burchfield et al., 2006), are a relatively new phenomenon in developing countries. As this urban development process is directly associated with various aspects of human life,¹ understanding its relationship with these aspects is imperative when examining whether urban areas remain liveable amid continuous improvements in transport networks.

The discussion on the effect of transport development on urban sprawl has advanced considerably in the last few decades. Recent literature seeks to provide more precise urban sprawl measurements and aims to deliver more rigorous empirical strategies to examine the causes of urban sprawl. Brueckner and Fansler (1983) pioneer the discussion to find a negative relationship between transport and urban land area. McGrath (2005) measures urban sprawl as the total urbanised land area and uses the consumer price index for private transport as a proxy for transport costs. The study finds that they are negatively correlated with spatial sizes of US metropolitan regions and confirms that fundamental economic factors such as transport costs can directly impact urban scale. A more sophisticated measurement that Burchfield et al. (2006) propose uses detailed spatial data to calculate the percentage of undeveloped land surrounding residential land (the sprawl index) as an indicator for

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¹ A wealth of literature has documented the complex relationship between urban sprawl and the urban environment (Gordon et al., 1989; Levinson and Kumar, 1994; Glaeser and Kahn, 2010; Newman and Kenworthy, 1989), labour productivity (Fallah et al., 2011), local public goods provision (Solé-Ollé and Rico, 2008), crime rates (Fallah et al., 2011), social and economic mobility (Ewing et al., 2016), and inequality (Lee et al., 2018).

sprawling development. The study uses streetcar passengers as a proxy for transport infrastructure to find that sprawling development occurs predominantly in more car-centric cities, confirming private transport's role in explaining urban sprawl. However, these studies do not point out any causal relationship between a transport network and urban sprawl. [García-López \(2019\)](#) is the most recent researcher to seek causality between transport development and urban sprawl. By employing historical transport networks as an instrument, the study finds that highway stocks expand residential land areas, increase the fragmentation of urban lots and induce the development of undeveloped land surrounding residential land. This study, however, indicates that such a phenomenon only occurs between cities.

This paper examines how transport networks influence the dynamics in land use within an urban area, distinguishing between urban expansion in general and urban sprawl in particular. We define urban expansion as the increase in built-up land observable with satellite data. This includes expansion in residential land, commercial areas and other human settlements. Urban expansion is indicated by changes in land use from a previously non-urban into an urban area. In contrast, we define urban sprawl as scattered developments in urban areas. An increase in urban sprawl thus means that the number of isolated developments on urban land has increased. To measure this condition, we use the sprawl index that [Burchfield et al. \(2006\)](#) introduce.

We study the Jakarta Metropolitan Area (JMA), one of the fastest-growing urban areas in Asia. Over the past 30 years, the JMA has experienced a substantial increase in population and urban areas, especially in suburban areas. As in many other metropolitan areas in Southeast Asia, spatial planning did not control the urban expansion process. Instead, it seems to have been guided by the presence of existing infrastructure. We construct urban land expansion and sprawl indices, which [García-López \(2019\)](#) introduces, using high-resolution spatial data from the Global Human Settlement Layer (GHSL) framework. This newly released dataset documents the changes in land-use patterns within metropolitan areas worldwide that were obscured in the coarser, regional-level statistics hitherto available for this type of research. To address potential non-random processes of improving highway access, we exploit aspects of the historical transport infrastructure that the colonial rulers built in the early 20th century as potential instrumental variables. We assume that the historical transport network is strongly associated with the transport network today but is less likely to correspond with today's urban land use ([Baum-Snow, 2007](#); [Duranton and Turner, 2012](#)).

The present study aims to explore the causal effect of highway expansion on urban sprawl within cities. Previous studies use the population density and economic activity in each region to investigate the impact of transport development within cities ([García-López, 2012](#); [Yudhistira et al., 2019](#)). In urbanised agglomerations of developing countries where land-use planning is less binding, such as the JMA, the built-up area is likely to increase, particularly in city suburbs, as highways expand. However, such improvements increase the sprawl index and hinder development toward a more compact structure that could otherwise be created.

This paper contributes to the existing literature on estimating the relationship between transport infrastructure and urban land use. We provide empirical evidence on how transport infrastructure shapes urban land use within cities, focusing specifically on the scattered development of urban land (sprawl) and urban land expansion. The results improve our understanding of how transport development affects urban sprawl differently within cities and between cities. While transport-induced urban sprawl is studied between cities in Europe ([García-López, 2019](#)) and between cities in the US ([Burchfield et al., 2006](#)), little is known about its effect within metropolitan areas in developing countries. We show that transport development still negatively affects urban land convergence, causing a slower convergence rate in regions with better transport development. Our study is also of interest internationally because the JMA, like other megacities in

developing countries, is a fast-changing urban structure ([World Bank, 2015](#)). The outcomes of this study contribute to an understanding of the externalities associated with urban sprawl led by highway expansion. Finally, our analysis contributes to the literature that explores the potential for using satellite data in examining urban sprawl phenomena. Our measurements follow recent satellite data applications that yield urban sprawl indicators such as urban land expansion ([Ahrens and Lyons, 2019](#); [Deng et al., 2008](#); [Oueslati et al., 2015](#)), the sprawl index ([Burchfield et al., 2006](#)) and the fragmentation of urban land ([García-López, 2019](#)). The data also allows us to use multiple urban sprawl indicators instead of using a single measurement, as previous studies do ([Brueckner and Fansler, 1983](#); [Burchfield et al., 2006](#); [McGrath, 2005](#)).

The remainder of this paper is organised as follows: [Section 2](#) provides background information on urban land use and the development of highways in the JMA. [Section 3](#) describes the urban sprawl measurements and sources of data used in this study. [Section 4](#) introduces our identification of the strategy, and [Section 5](#) discusses the results. Finally, the last section concludes the study.

2. Context: land and highway development in the Jakarta Metropolitan Area

2.1. The land sprawling process

Indonesia has the third-largest amount of urban area in East Asia, and this is growing rapidly. Its urban area growth rate of 1.1% per year between 2000 and 2010 was second only to mainland China's. Jakarta is the most densely populated metropolitan area in Indonesia; with only 12% of the country's built-up land, it houses approximately 20% of its urban population. In addition, the JMA ranks as the second-densest urban area in East Asia ([World Bank, 2015](#)). The JMA experienced an annual population growth rate of 2.8% between 2000 and 2010.² Moreover, urban density in the JMA increased from 12,200 persons/km² of urban land in 2000 to more than 14,600 in 2010. This indicates that the benefits of agglomeration in the JMA are still attractive enough to attract more people to live and work in the area. The development of the built-up area in the JMA can be seen in [Fig. 1](#). The green area depicts the new built-up area developed between 1990 and 2015.

The growth, however, differs throughout the JMA. The city of Jakarta's population grows more slowly than its surrounding suburbs, and the share of JMA's population living in the city of Jakarta has decreased by 5%.³ The trend of decreasing population density in the centre of Jakarta started in the 1970s. During the 1990s, Jakarta entered its suburbanisation stage, while similar metropolitan areas such as Metro Manila and Bangkok were still in a pre-suburbanisation stage ([Murakami et al., 2005](#)). The urban development of the JMA is also characterised by a typical suburbanisation sprawl along its development corridors ([Goldblum and Wong, 2000](#)). This process is common in Southeast Asian and North American megacities ([Silver, 2007](#)), possibly because of the emergence of suburban housing markets led by the growing affluence of the suburban population and the increasing importance of the private sector in the land development process ([Winarso and Firman, 2002](#); [Winarso et al., 2015](#)). Spatial regulation in the JMA is also considered inconsistent and fragmented since it includes various authorities with different ideas and priorities ([Hudalah and Woltjer, 2007](#)). Therefore, spatial regulation within the JMA may be less binding than in other metropolitan areas within Southeast Asia. These less-binding regulations may further increase the sprawling and dispersed development of the JMA and foster the inefficient provision of

² Author's calculation using population census data from the National Bureau of Statistics (BPS, 2010). Population growth calculated using the compound annual growth rate for the 10 years.

³ Author's calculation using population census data from the National Bureau of Statistics (BPS, 2010).

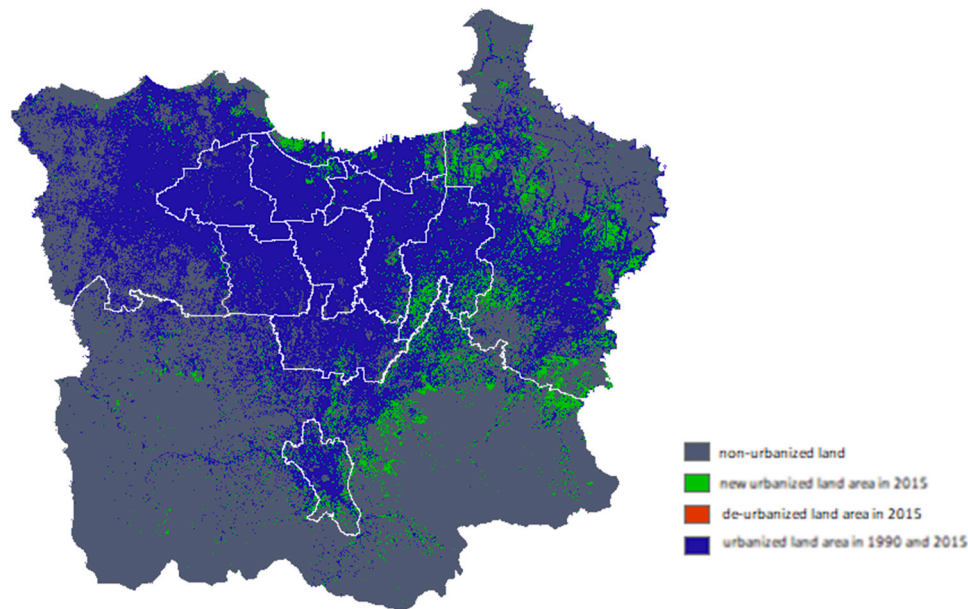


Fig. 1. Land-use changes in the JMA in 1990–2015.

Source: The Global Human Layer Settlement, [European Commission \(2018\)](#).

essential services within the area ([Silver, 2007](#)).

2.2. Highway development

Extensive highway development in the JMA started with the construction of the Jagorawi toll road in 1973, which was used to improve the connection between Jakarta and Bogor in the southeast. The effort to connect Jakarta and its periphery continued with the development of new toll roads to Tangerang (1984) and Cikampek (1988), as well as the development of the Jakarta Inner Ring Road (1989). Since 1990, no less than 150 km of highway have been built in the area. The national highway authorities plan to further expand the highway networks inside the JMA until 2030, notably through the construction of the second

layer of the Jakarta Outer Ring Road (JORR II). [Fig. 2](#) depicts the highways that existed in 1989 in green, the current highway rays in blue and the expansion plan until 2030 in red.

Improvements in transport infrastructure in the JMA over the last three decades have occurred in line with the economic and land-use development that has taken place there. This has resulted in an increase in industrial estates and residential towns in the suburbs ([Hudalah et al., 2013](#)), industrial and population decentralisation ([Henderson et al., 1996](#)) and the transformation of rural land to new towns in the peri-urban area of Jakarta. Over 300,000 ha of rural land was transformed into new towns, mostly located in the peri-urban area of Jakarta, causing the urban area of the JMA to expand from 10 km² in 1970–40 km²–45 km² in 2015 ([Fitriyanto et al., 2019](#)). The

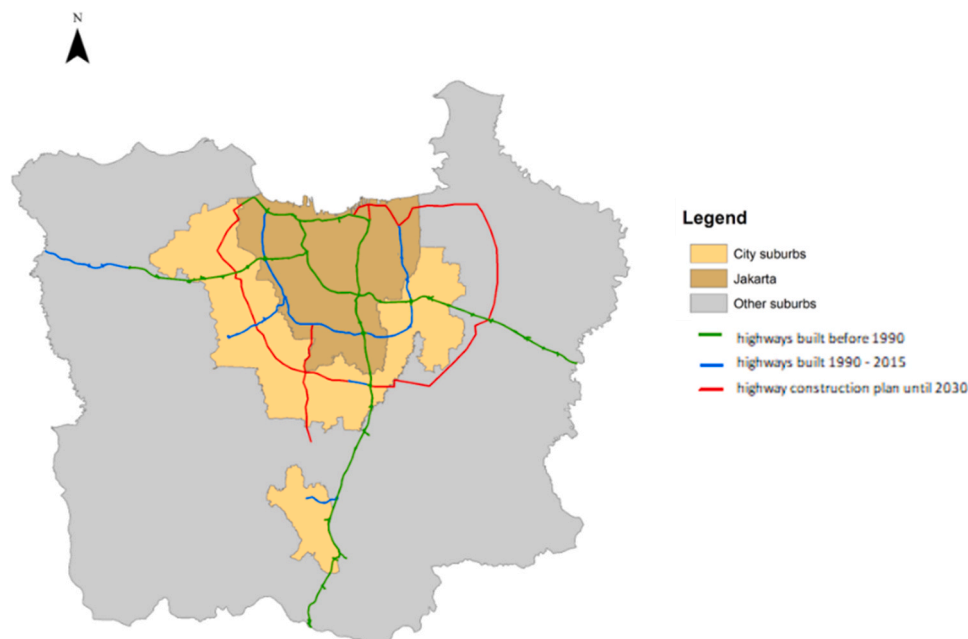


Fig. 2. Highway development in the JMA.

Source: [National Highway Authority \(2019\)](#).

development of the JMA was influenced not only by the toll road network but also by the revitalisation of the railway network. The Dutch constructed most of the railway lines in Indonesia, including the JMA railway network, during the colonial era. The JMA railways were abandoned after Indonesia's independence in 1945 but were then revived in the 1970s by the national railway corporation (PNKA). Since the early 1990s, the national railway authority has gradually reopened old lines, such as Tanah Abang–Serpong (1992), Jakarta–Bekasi (1992), Duri–Tangerang (1997) and Serpong–Parung Panjang (2009).

3. Data construction

3.1. Measuring urban sprawl

A wealth of literature discusses ways to measure urban development and, specifically, urban sprawl. In this paper, we quantify urban development using two indicators: the sprawl index and an urban land expansion metric. The first applied sprawl index was introduced to describe how often residential development occurs more than a certain distance away from existing urban developments (Burchfield et al., 2006). The index is generated by calculating the share of open space in the direct surroundings of a built-up cell and then averaging the values for all built-up cells per unit of analysis. An increase in the sprawl index over time can be interpreted as an increase in built-up cells isolated from other artificial land, indicating the more scattered development of residential areas. Ahrens and Lyons (2019) and Garcia-López (2019) employ a similar indicator. The second indicator is urban land expansion, which is defined as the increase in the urban area in a particular region as a fraction of the initial urban area. It indicates that a region is experiencing urban growth, irrespective of the pattern.

The difference between the sprawl index and urban expansion is shown in Fig. 3. Each cell represents a 1 km² area, and T0 is the initial condition of a district. Urban land expansion is defined as the increase in built-up cells in a particular year since the base year. In T1-A, there is a 5 km² increase in built-up cells compared to T0 (urban expansion equals 0.5), while in the alternative development option T1-B, urban expansion equals 0.3 (3 km²). To determine the sprawl index, we first calculate the percentage of open space in the eight cells directly neighbouring each built-up cell. The sprawl index for each district is calculated by averaging the numbers for every built-up cell within a district. As shown in Fig. 3, the sprawl index for T1-A is lower (0.34) than that of T1-B (0.48), indicating a higher degree of the sprawling development of urban areas in T1-B. Fig. 3 illustrates that substantial urban expansion can result in a lower sprawl index. However, even with less expansion, the sprawl index can also increase.

3.1.1. Data

In this study, we use the GHSL data from the Joint Research Centre of the European Commission. This data is derived from the Landsat image collection available for the years 1990, 2000 and 2014 at a spatial resolution of 38 m. A cell contains a value of one if it is considered built-up and, otherwise, zero. With regard to transport development, different proxies of transport costs are discussed in the literature that examines the impact of transport on urban development. They include the percentage of people using public transport and the percentage of people owning automobiles (Brueckner and Fansler, 1983), as well as the length of highways (Garcia-López, 2019; Oueslati et al., 2015) and time spent on commuting (Ahrens and Lyons, 2019). In this study, we use change in distance to the nearest highway ramp as a measure of transport improvement.

To control for exogenous distance variables (distance to the nearest railway station, district centres and coastline), we use topographic data from OpenStreetMaps (OSM) as a baseline. The data is then cross-checked with various sources, such as the National Highway Authority of Indonesia, KAI Commuter Jabodetabek (the railway authority) and the Indonesian Geospatial Information Agency (BIG). This ensures that there is no significant difference causing measurement bias in our calculations. To calculate geographical variables (elevation and the terrain ruggedness index), we employ official digital elevation maps data from BIG using the finest resolution of 12 m² spatial units. Lastly, to avoid the modifiable areal unit problem that arises due to substantial differences among administrative areas in Jakarta, city suburbs and other suburbs, we divide our unit of analysis into identical 4 km² areas (2 km × 2 km spatial units), as others advocate (Jacobs-Crisioni et al., 2014). A more detailed description of each variable with key statistics is presented in Appendix 1.

4. Identification strategy

Based on previous empirical studies that evaluate the effects of infrastructure development (Baum-Snow, 2007; Garcia-López, 2019; Yudhistira et al., 2019), we estimate the effect of highway expansion on urban sprawl as follows:

$$\Delta \ln(y_i) = \alpha_0 + \alpha_1 \Delta(\text{dist.to.ramps})_i + \beta x_i + \epsilon_i \quad (1)$$

where y_i refers to two variables capturing the degree of sprawling development: (a) the percentage of undeveloped land surrounding the built-up cell (sprawl index), implying a scattered development, and (b) the built-up area (in km²), indicating urban land expansion. Our main variable of interest is the change in distance to highway ramps from 1990 to 2014, indicating an improvement in access to transport in the JMA. x_i is a set of control variables consisting of initial developments,

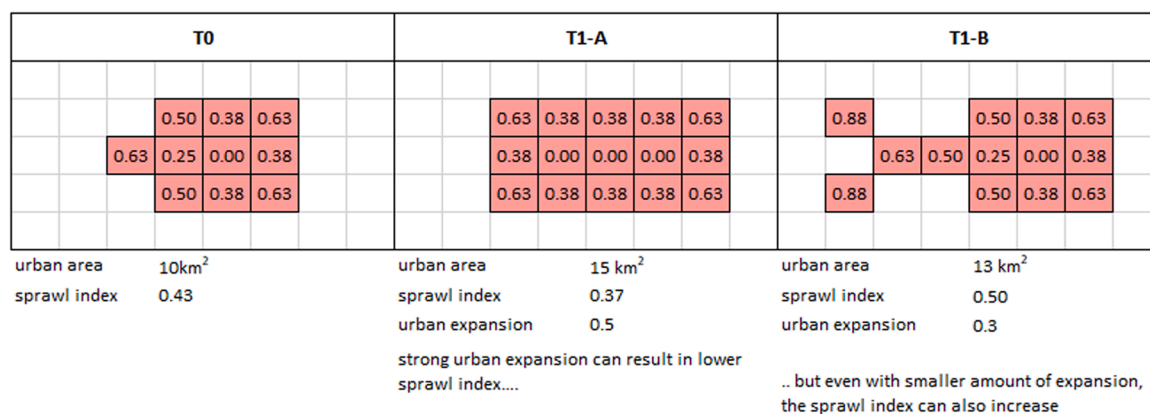


Fig. 3. Illustration of the urban expansion and sprawl index calculations. Note: red cells indicate the built-up area and white cells the non-built-up area. The number inside each built-up cell represents the percentage of undeveloped cells surrounding the built-up cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

distance (distance to railway stations, district centres and coastlines) and geographical variables (elevation and terrain ruggedness index). As the regression is expressed as the difference in the log, γ represents percentage point changes in the degree of sprawling development associated with a 1 km change in the distance to highway ramps. The descriptive statistics of these variables are presented in [Appendix 1](#).

The difference in log expression enables us to control for unobserved time-invariant heterogeneity that may correlate with the highway and urban sprawl expansion. We include some time-invariant variables and the initial land development or urban sprawl index as controls to account for the possibility that these variables may correlate with the improvements in highways and determine the change in urban sprawl.

Estimating [Eq. \(1\)](#) using the ordinary least squares (OLS) may still produce biased results due to the reverse causality problem: It is plausible that urban development fosters highway expansion. To address this issue, we adopt an instrumental variable estimation method using historical roads and railways from [Fig. 4](#) as potential instruments. Historical transport infrastructure, such as roads and railways are commonly used as instruments since they are exogenous to today's urban land use, but strongly associated with current transport network development ([Baum-Snow, 2007](#); [Duranton and Turner, 2012](#)).

To account for this identification issue, we use the distance to historical transport to predict the improvement in highway access at the first stage and the predicted value of the improvement to estimate the effect on urban sprawl at the second stage, as follows:

$$\Delta \ln(\text{dist.to ramp}_i) = \gamma_0 + \gamma_1(\text{dist.to hist.road})_i + \theta x_i + u_i \quad (2)$$

$$\Delta \ln(y_i) = \phi_0 + \phi_1 \Delta \ln(\text{dist.to ramp}_i) + \pi x_i + \epsilon_i \quad (3)$$

where $\Delta \ln(\text{dist.to ramp}_i)$ is the predicted change in highway access. Since these instruments may not be placed randomly due to the influence of other factors, it is important to control for initial urban development and geographical variables as well ([García-López, 2012](#)). Thus, we prefer to use the full-specification model to control for initial urban development, distance and geographical variables. In this paper, by controlling for the distance to the nearest railway station, we acknowledge the role of a railway network in shaping urban forms, but

we ignore any potential for railways to induce urban sprawl. We consider the small statistical significance in [Yudhistira et al.'s \(2019\)](#) recent study as the main reason to ignore this causal impact.

Relevance and exogeneity conditions should be met for our instruments to be valid. Under the relevance condition, the instruments used in the second-stage estimation should not be weak. The Kleibergen–Paap F-statistic of instruments used in two-stage least squares (TSLS) regressions need to exceed the size and relative bias critical value to reject the null hypothesis of weak instruments ([Stock and Yogo, 2005](#)). For an instrument to be suitable for our TSLS estimation, the first-best option is to choose an instrument that satisfies not only the validity criteria but also the following conditions: (1) It significantly affects the endogenous variables (first stage), and (2) it significantly affects the outcome variables (reduced form). Previous studies, however, do not always provide the first-best instruments for analysis due to insignificant results for the reduced-form estimations ([García-López, 2012](#); [Yudhistira et al., 2019](#)). To overcome this problem, we prioritise the first-stage estimation results in choosing which instruments to use. In a less restrictive way, [Yudhistira et al. \(2019\)](#) choose the instruments by looking at which instruments produce stronger F-statistics in the second-stage estimations.

[Fig. 5a](#) and [b](#) depict the relationship between the distance to historical roads and changes in distance to the nearest highway ramps, also indicating our first-stage relationship between these variables. The figures indicate that areas nearer historical roads experience greater improvement in highway access. In [Fig. 6](#), we show the correlation between the distance to historical roads and our urban development indicators—urban expansion and the sprawl index—as an uncontrolled association between these variables. We find that greater urban expansion and more scattered development are located in areas that experience greater improvement in highway access.

The exogeneity condition requires that the instruments do not directly affect the outcome variables but are channelled through the endogenous variables. In this case, we use transport infrastructure built a century or more ago, and we argue that it is unlikely that this was built in anticipation of the current urban spatial patterns ([García-López, 2012](#)). To check for the validity of this assumption, we employ a similar approach to [Aladangady \(2017\)](#) to check whether other determinants of

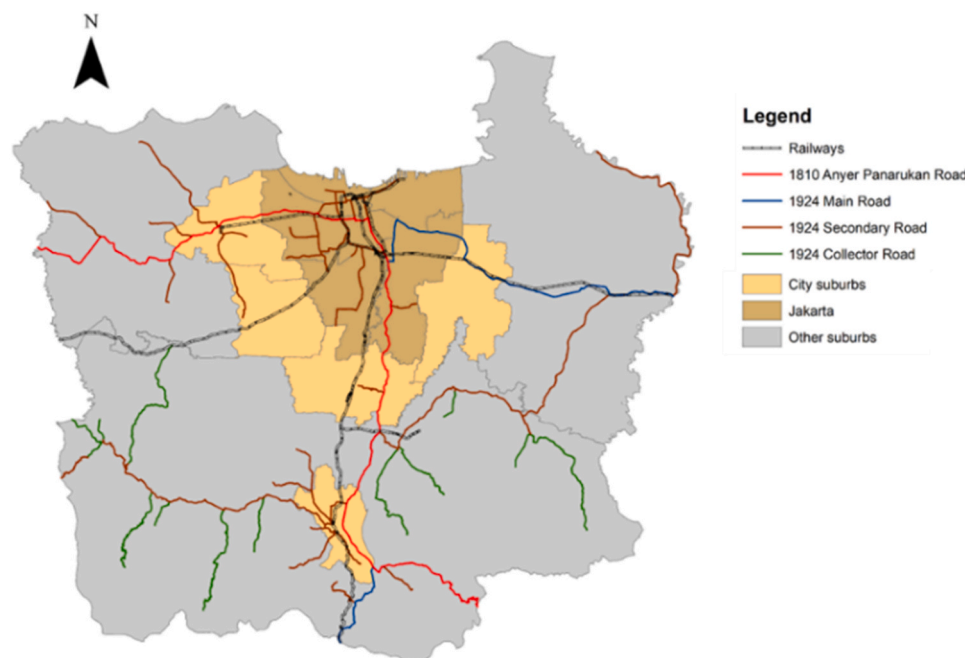


Fig. 4. Historical roads in the Jakarta Metropolitan Area.
Source: Dutch Colonial Maps—Leiden University Library (2019).

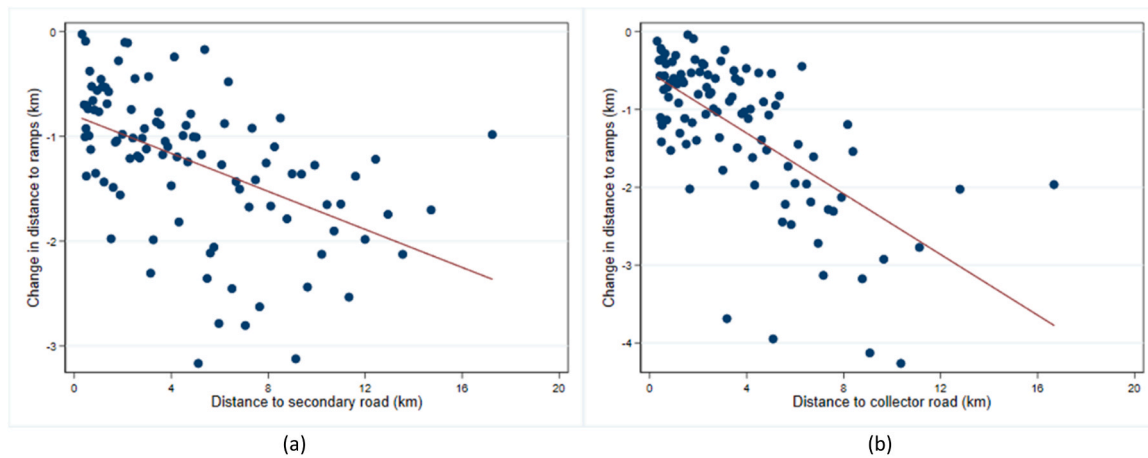


Fig. 5. Distance to historical roads after improvement in access to highway ramps.

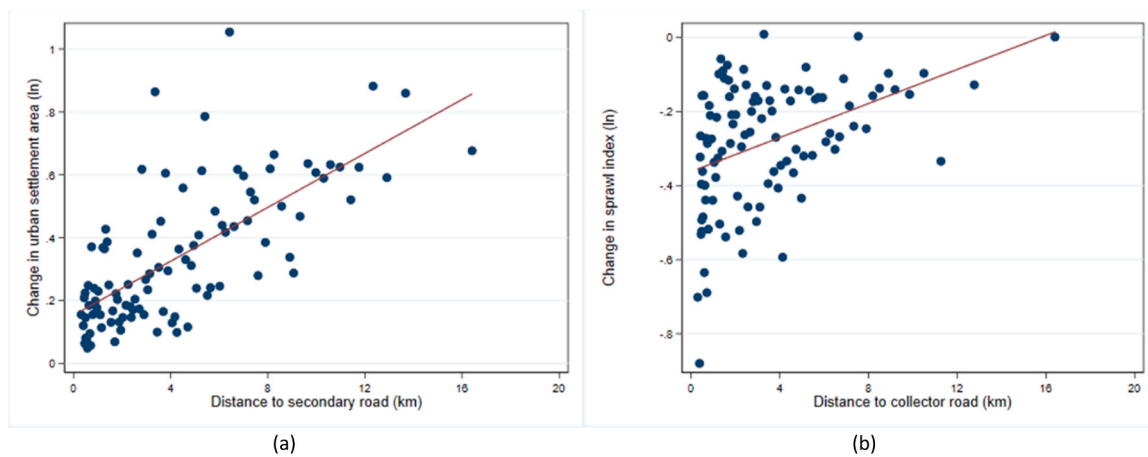


Fig. 6. Changes in urban sprawl indicators and distance to historical roads.

urban forms are significantly correlated with the instruments at the sub-district level. We use non-transport infrastructure data for the year of 2000 from the National Village Potential Survey as other potential channels of historical roads to affect changes in urban forms from 1990 to 2014.

In Table 1, we regress other determinants of urban forms, including number of primary schools, district health centres, markets, and poor households in the year of 2000 on distance to historical roads. These variables represent non-transport facilities located in each sub-district. The presence of such facilities may correspond to different urban forms (Zhang, 2001; De Vos and Witlox, 2013), however, we expect that it is exogenous to where historical roads are located and thus bears no significant relation with them. Our estimated coefficients in columns (1) and (2) in Table 1 suggest that exogeneity of the instruments is indeed a plausible assumption.

5. Results and analysis

5.1. Ordinary least squares estimation

For descriptive purposes, we present the OLS results for urban expansion and the sprawl index in Table 2. Column (1) shows a negative correlation between changes in distance to the nearest highway ramps and changes in the sprawl index. The magnitude, however, becomes smaller as we control for distance and geographical variables, as shown in Columns (2) and (3). Our preferred specification in Column (3) indicates that a 1 km improvement in access to a highway corresponds to a

Table 1
Effect of excluded instruments on sub-district level variables.

Instruments:	Distance to 1924 Secondary Road (1)	Distance to 1924 Collector Road (2)	Number of sub-districts (3)
Number of primary schools	0.028 (0.111)	-0.101 (0.070)	106
Number of district health centres	-0.074 (0.066)	-0.139 (0.095)	104
Number of markets	-0.121 (0.090)	-0.076 (0.073)	106
Number of poor households	-0.070 (0.068)	0.049 (0.118)	106

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Each row represents a separate regression of determinants of urban forms on historical roads. Estimation is in natural logarithm form conditional on geographic and distance variables. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index. See Appendix 1 for a more extensive description of these variables.

3.8% increase in scattered development within the JMA. We provide a similar estimation for urban land expansion presented in Columns (4), (5) and (6). Since we define changes in distance to nearest highway

Table 2

OLS Estimation—Jakarta Metropolitan Area.

Dependent variable	Δ ln (sprawl index)			Δ ln (built-up area)		
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)
Δ distance to nearest highway ramp	– 0.048*** (0.004)	– 0.046*** (0.004)	– 0.038*** (0.004)	0.054*** (0.004)	0.026*** (0.004)	0.016*** (0.004)
ln (sprawl index – 1990)	N	Y	Y	N	Y	Y
ln (built-up area – 1990)	N	Y	Y	N	Y	Y
Distance	N	Y	Y	N	Y	Y
Geography	N	N	Y	N	N	Y
Observations	1421	1421	1421	1421	1421	1421
Mean of dependent variables	– 0.282	– 0.282	– 0.282	0.342	0.342	0.342
Adjusted R ²	0.048	0.309	0.332	0.041	0.495	0.510

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index. See [Appendix 1](#) for a more extensive description of these variables.

ramps as distance to nearest ramps in 2015 minus the distance in 1990, the areas experiencing the biggest improvement has a larger negative value than those who experienced smaller improvement. Our preferred OLS result controlling for distance and geographical variables produces a positive correlation between changes in the distance to highway ramps (2015–1990) and new, built-up areas. This implies that improvement in access to the nearest highway ramps is associated with a low level of urban expansion in the JMA. A 1 km improvement in access to the nearest highway ramp corresponds to a 1.6% reduction in urban expansion within the JMA. These results, however, may well be invalid due to a reverse causality bias. Therefore, we present the instrumented variable approach in the next section.

5.2. Two-stage least squares estimation

5.2.1. First-stage estimates

The first-stage OLS estimates the effect of proximity to historical infrastructure on improvement in highway access (see [Table 3](#)). Columns (1) and (2) in [Table 3](#) include the sprawl index for 1990, representing the first-stage estimation for the sprawl index. Meanwhile, Columns (3) and (4) use the initial built-up area in 1990 as a control for the initial development of urban settlement and thus show the first-stage estimation result for urban expansion. Our results show that the 1924

Table 3

First-stage estimation.

Dependent variable	First-stage estimates Δ distance to the nearest highway ramp			
	(1)	(2)	(3)	(4)
Distance to 1924 secondary road	– 0.128*** (0.022)		– 0.178*** (0.024)	
Distance to 1924 collector road		– 0.188*** (0.030)		– 0.213*** (0.029)
ln (sprawl index—1990)	Y	Y		
ln (built-up area—1990)			Y	Y
Distance	Y	Y	Y	Y
Geography	Y	Y	Y	Y
Observations	1421	1421	1421	1421
Mean of dependent variables	– 1.239	– 1.239	– 1.239	– 1.239
Adjusted R ²	0.236	0.243	0.243	0.249

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index.

secondary and collector roads are statistically significant in explaining the variation in highway access improvement. The results statistically satisfy the correlation condition, thus allowing its application as an instrumental variable. The exogeneity condition for these historical roads is met, as they are unlikely to directly affect urban development. Meanwhile, when new highway rays are built, these roads are considered to take into account accessibility to the highway ([García-López, 2012](#)). Since two of these conditions are met, we will use these variables as instruments. The estimated coefficients indicate that proximity to historical roads corresponds to a smaller improvement in the distance to the nearest highway ramp. This suggests that newly built highways tend to be located away from existing road networks, possibly for economic or political reasons ([García-López, 2012](#)).

5.2.2. The effects of transport improvement on urban form

We use the distances to 1924 secondary and collector roads separately as instruments. When we control for initial urban development, distance and geographical variables, our selected instruments produce high F-statistics in all specifications, indicating that our instruments are not weak. In [Table 4](#), Columns (1) and (2) present the effects of an improvement in highway access in the sprawl index within the JMA. On average, areas in the JMA have been experiencing a lower sprawl index throughout 1990–2014 (see [Appendix 1](#)), indicating that new land developments in the JMA tend to converge with the initial developments. However, after controlling for the exogenous variables related to land

Table 4

The effect of transport development on urban sprawl.

Dependent variable	Δ ln (sprawl index)		Δ ln (built-up area)	
	TSLs (1)	TSLs (2)	TSLs (3)	TSLs (4)
Δ distance to the nearest highway ramp	– 0.066** (0.029)	– 0.096*** (0.032)	– 0.032 (0.024)	– 0.009 (0.020)
ln (sprawl index—1990)	Y	Y		
ln (built-up area—1990)			Y	Y
Distance	Y	Y	Y	Y
Geography	Y	Y	Y	Y
Observations	1421	1421	1421	1421
Mean	– 0.282	– 0.282	0.342	0.342
Kleibergen–Paap F-statistic	38.83	38.55	56.02	53.99
Instruments				
Distance to 1924 secondary road	✓		✓	
Distance to 1924 collector road		✓		✓

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index.

use, the estimated coefficients show that a 1 km improvement in highway access increases the sprawl index by 6.6–9.6%. It further indicates that highway expansion hinders the process of more compact development within the JMA.

Our preferred OLS coefficient provides a slight underestimation compared to its TSLS counterpart, largely due to not controlling for the reverse causality bias. In terms of magnitude, by comparing the magnitude of the means, our estimated coefficients have a considerably high magnitude. Thus, it is important to be meticulous in interpreting the results. In examining the effect on urban land expansion, our TSLS coefficients show no statistical evidence of transport-led urban land expansion in the JMA. Our estimated coefficient provides a different effect than its OLS counterpart, indicating an overestimated result in our OLS coefficient due to a reverse causality bias.

To further examine the effect of transport development on urban land expansion, we perform a similar approach at the sub-sample level. We divide the JMA into three sub-samples (Jakarta, city suburbs and other suburbs) by considering their administrative category. Table 5 presents the sub-sample results for urban expansion. We use distance to the historical 1815 Anyer–Pasarukin road as an instrument for Jakarta. In contrast, for city suburbs and other suburbs, we select the distance to the old railway station and the 1924 secondary roads. All of our instruments are not considered weak according to their high Kleibergen–Paap F-statistics. Our estimated coefficients for Jakarta and other suburbs are not statistically different from zero, indicating that urban expansion is not caused by highway expansion. Our estimates for city suburbs, however, show a negative effect that is statistically different from zero. A 1 km improvement in the distance to a highway ramp induces the expansion of urban areas by 1.6%. It confirms the existence of transport-led urban expansion within the city suburbs. To complement the analysis, we also add sub-sample results for the sprawl index in Appendix 4.

5.2.3. Robustness check

To check the consistency of our estimations, we perform similar estimations using different spatial units, samples and datasets (see Tables 6 and 7). In Table 6, Columns (1) to (5) present the estimation results when adding one district adjacent to the JMA to the observations. We estimate the effect of improvements in highway access on urban sprawl using the same instrument (the 1924 collector roads) as our main estimation in Column (2) of Table 3. The addition of one district located adjacent to the JMA produces a result consistent with our main estimation. The only opposite effect is observed when JMA is combined with the Serang District. The estimated coefficient results in a positive

sign but with a low Kleibergen–Paap F-statistic value, which indicates that the instrument is weak for explaining the effect in this sub-sample. The impact of the distance to the nearest highway ramp on the regressions for the JMA and surrounding districts varies from 8% to 12%, implying a slightly larger effect than that of our main result in Column (2) of Table 3. The larger effect may be caused by a combination of low land value in surrounding districts and the absence of strict land conversion regulations. As such, it is easier to achieve land conversion in the surrounding districts of the JMA. Columns (6) to (8) show our estimated coefficients when we limit our JMA sample area to 30 km, 40 km or 50 km from central Jakarta. Our results indicate that the effect is negligible and statistically insignificant within a radius of 30 km but evident within 40 km (4.7%) and even larger within 50 km (10.2%). As an additional robustness check, we apply a similar approach as Garcia-López (2019) and reduce our sample by excluding the bottom 5% and 10%, as well as the top 5% and 10%, of our observations (see Table 6, Columns [9] to [12]). The results are robust for the top 5% and 10% exclusion but become insignificant when excluding the bottom 5% and 10% of the observations.

In addition to changing the number of observations, we check the robustness of our results by using different spatial units and data sources. The main result using a 2 km × 2 km spatial unit is repeated in the first column of Table 7, while Column (2) shows the impact of using larger 3 km × 3 km spatial units and fewer observations. Our estimated coefficient in Column (2) is consistent with our main TSLS results and still indicates a negative relationship between improvement in highway access and the sprawl index. The magnitude of the coefficient is lower than that of our main result in Column (1) and only statistically significant at the 10% level. Therefore, we prefer the results obtained at the more detailed level.

Finally, we test the impact of using an alternative dataset for capturing urban development. For this dataset that the European Space Agency's Climate Change Initiative (ESA CCI) provided, we also apply 2 km and 3 km spatial units. Again, the results are consistent with our main result, but the magnitude is substantially higher than that of our preferred estimation. This stronger impact may be related to the different resolutions of the data. A larger resolution overestimates the existence of small open spaces; it may neglect the local spatial variation caused by urban green space, rivers, etc. Our main estimation uses the GHSL data of 38 m spatial resolution, where the ESA CCI data is provided at 300 m spatial resolution.

We also perform numerous robustness checks related to urban expansion, as shown in Appendix 5. Most of the specifications produce no statistical evidence of transport-led urban expansion in the JMA.

Table 5
The effect of changes in transport access and urban land expansion in JMA regions.

Dependent variable	Δ ln (built-up area)					
	Jakarta		City suburbs		Other suburbs	
	OLS (1)	TSLS (2)	OLS (3)	TSLS (4)	OLS (5)	TSLS (6)
Δ distance to nearest highway ramp	0.005 (0.004)	0.011 (0.010)	0.002 (0.002)	− 0.016*** (0.005)	0.027*** (0.006)	0.037 (0.040)
ln (built-up area–1990)	Y	Y	Y	Y	Y	Y
Distance	Y	Y	Y	Y	Y	Y
Geography	Y	Y	Y	Y	Y	Y
Observations	155	155	215	215	1051	1051
Mean	0.066	0.066	0.106	0.106	0.431	0.431
Adjusted R ²	0.883		0.842		0.503	
Kleibergen–Paap F-statistic		19.43		30.08		29.55
Instruments						
Distance to 1815 Anyer–Pasarukin road		✓				
Distance to old railway station				✓		
Distance to 1924 secondary road						✓

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index.

Table 6
Robustness of other empirical specifications with different study areas and the number of observations.

Dependent variable		$\Delta \ln$ (sprawl index 1990–2014)											
Category	Adjacent districts				Central Jakarta radius				Exclusion of top and bottom percentiles				
Sample	JMA	JMA	JMA	JMA	JMA	JMA	30 km	40 km	50 km	Top 5%	Top 10%	Bottom 5%	Bottom 10%
Method	TSLs (1)	TSLs (2)	TSLs (3)	TSLs (4)	TSLs (5)	TSLs (6)	TSLs (7)	TSLs (8)	TSLs (9)	TSLs (10)	TSLs (11)	TSLs (12)	TSLs (12)
Δ distance to nearest highway ramp	– 0.126***	– 0.095***	– 0.081***	– 0.098**	0.309**	– 0.002	– 0.047**	– 0.102***	– 0.101***	– 0.106***	– 0.025	– 0.019	
$\ln$ (sprawl index—1990)	(0.031)	(0.034)	(0.027)	(0.049)	(0.143)	(0.029)	(0.022)	(0.028)	(0.033)	(0.034)	(0.021)	(0.020)	
Distance	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Geography	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	1477	1430	1587	1498	1523	504	866	1241	1393	1337	1346	1271	
Mean	– 0.272	– 0.280	– 0.281	– 0.268	– 0.267	– 0.531	– 0.399	– 0.321	– 0.288	– 0.300	– 0.228	– 0.200	
Kleibergen–Paap F-statistic	46.51	34.53	52.65	13.76	5.93	57.93	84.94	48.22	37.002	35.51	33.35	33.24	

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index. All specifications use the distance to the 1924 collector road as an instrument.

Table 7
Robustness of other empirical specifications: data sources and spatial units.

Dependent variable	$\Delta \ln$ (sprawl index 1990–2014)			
Category	Spatial unit		Data source	
Sample	2 km	3 km	ESACCI 2 km	ESACCI 3 km
Method	TSLs	TSLs	TSLs	TSLs
	(1)	(2)	(3)	(4)
Δ distance to nearest highway ramp	– 0.096*** (0.032)	– 0.066* (0.037)	– 0.348*** (0.084)	– 0.295*** (0.097)
$\ln$ (sprawl index—1990)	Y	Y	Y	Y
Distance	Y	Y	Y	Y
Geography	Y	Y	Y	Y
Observations	1421	635	703	367
Mean	– 0.282	– 0.236	– 1.055	– 1.117
Kleibergen–Paap F-statistic	38.55	18.41	37.84	24.60

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index. All specifications use the distance to the 1924 collector road as an instrument.

Only in the specifications that combine the JMA with neighbouring districts (Lebak or Serang, both located on the western side of the JMA) do we find that a 1 km improvement in highway access leads to 6.8–9.9% of the urban land expansion.

5.2.4. Falsification test

We use a falsification test to further confirm that our results cannot be attributed to chance. This test is performed by randomising all of the observations of our instrumental variables so that each observation is now attributed to different instruments' observations in our TLS estimation. A statistically significant result in a falsification test would indicate that our previous estimation suffered from placebo effects and, consequently, that we should reject our interpretation in the previous section. [Table 8](#) presents our falsification test results for both the sprawl index and urban land expansion. The results show that none of the estimates is statistically different from zero, refuting the possibility of placebo effects in our identification strategy.

6. Discussion

In this study, we estimate the effects of improved access to highways on urban sprawl phenomena in the JMA. While such studies are common in developed countries (e.g., [García-López, 2019](#)), limited evidence is available for fast-developing metropolises. Urban sprawl is characterised by expansion in developed land area ([Deng et al., 2008](#); [McGrath, 2005](#)) and scattered developments ([Angel et al., 2012](#); [Burchfield et al., 2006](#)). Our study employs measurements of both features to provide a more comprehensive view of urban sprawl in the JMA.

Our results align with the general understanding of how transport costs affect urban sprawl (Brueckner and Fansler, 1983; Burchfield et al., 2006; McGrath, 2005; Young et al., 2016). We find that highway expansion induces the scattered development of urban land and urban land expansion within the JMA. The estimates also indicate that the magnitude is lower in the city centre, where the area is already more compact, than in the less-developed urban areas farther away from Jakarta. This may be because the development of new highways between 1990 and 2015 was concentrated on the fringe areas of Jakarta. Thus, city suburbs are likely to experience a stronger effect in land expansion.

Despite several studies that take a similar approach, we did not find any directly comparable studies. However, our estimates suggest that

Table 8
Falsification test.

Dependent variable	$\Delta \ln$ (sprawl index)				$\Delta \ln$ (built-up area)			
	JMA	Jakarta	City suburbs	Other suburbs	JMA	Jakarta	City suburbs	Other suburbs
	TSLS (1)	TSLS (2)	TSLS (3)	TSLS (4)	TSLS (1)	TSLS (2)	TSLS (3)	TSLS (4)
Δ distance to the nearest highway ramp	1.113 (1.790)	− 26.492 (1197.621)	0.681 (2.083)	− 0.182 (0.227)	− 0.514 (1.076)	− 0.206 (1.229)	0.151 (0.209)	0.277 (0.463)
$\ln$ (sprawl index—1990)	Y	Y	Y	Y				
$\ln$ (built-up area—1990)					Y	Y	Y	Y
Distance	Y	Y	Y	Y	Y	Y	Y	Y
Geography	Y	Y	Y	Y	Y	Y	Y	Y
Observations	1524	155	215	1154	1421	155	215	1051
Mean	− 0.296	− 0.828	− 0.419	− 0.202	0.342	0.066	0.106	0.431
Kleibergen–Paap F-statistic	0.387	0.000	0.107	0.789	0.265	0.027	0.534	0.558
Instruments								
Distance to Anyer–Pasarukin road		✓				✓		
Distance to old railway station							✓	
Distance to 1924 secondary road			✓	✓	✓			✓
Distance to 1924 collector road	✓	□						□

Note: robust standard error in parentheses. *, **, *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. Distance variables include distance to the nearest railway station, distance to central Jakarta, distance to the central district and distance to the coastline. Geographical variables include elevation and the terrain ruggedness index.

urban development within cities exhibits a similar pattern to that documented in other studies focusing on the space between cities (Deng et al., 2008; Garcia-López, 2019). Our estimated coefficients indicate that a 1 km improvement in access to highway ramps leads to a 6.6% increase in the sprawl index within the JMA and expands the urban land in city suburbs by 1.6%.

Recent advances in the provision of openly accessible, detailed spatial data, such as the GHSL data we apply in this study, hold great promise for extending the empirical literature on urban development in developing countries. Previous studies on urban growth in Jakarta, for example, apply relatively coarse data and only describe basic trends (Firman, 2004; Henderson et al., 1996; Hudalah and Firman, 2012).

7. Conclusion

By employing a detailed time series of spatial data capturing urban development, we investigate whether transport-induced urban sprawl is evident in the JMA. Our study uses two measurements of urban sprawl—the sprawl index and urban land expansion—and exploits the variations in historical transport infrastructure as instruments to overcome the potential endogeneity issue. We find that improvement in highway access results in the scattered development of new, built-up areas in the JMA: As highways expand, new urban land is more likely to become isolated from the existing urban land. This result is robust across all of our specifications. In addition, we find highway-led urban expansion in the city suburbs of the JMA, indicating that urban development in the JMA occurs in two forms: the scattered development of the urban area and urban land expansion.

Sprawling development is documented to have adverse impacts on, for example, energy consumption (Glaeser and Kahn, 2010; Proque et al., 2020), productivity (Fallah et al., 2011), social issues (Ewing et al., 2016; Solé-Ollé and Rico, 2008) and equality (Lee et al., 2018). When considering the development of new highways in urban areas such as the JMA, policymakers should be aware of the indirect impacts of their investments on urban development and, thus, incorporate them in their cost–benefit analyses.

Our study points out that private transport strongly influences the development of urban areas within a car-centric metropolitan area such as the JMA. This suggests that introducing policies to increase private

transport costs, such as fuel taxes and road pricing, may impact urban development by initiating changes in location choice behaviour. However, more comprehensive research related to transport policies and individuals' behaviour in the JMA is needed to better understand how different groups in society will adapt their travel and location behaviour.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [10.1016/j.landusepol.2021.105856](https://doi.org/10.1016/j.landusepol.2021.105856).

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